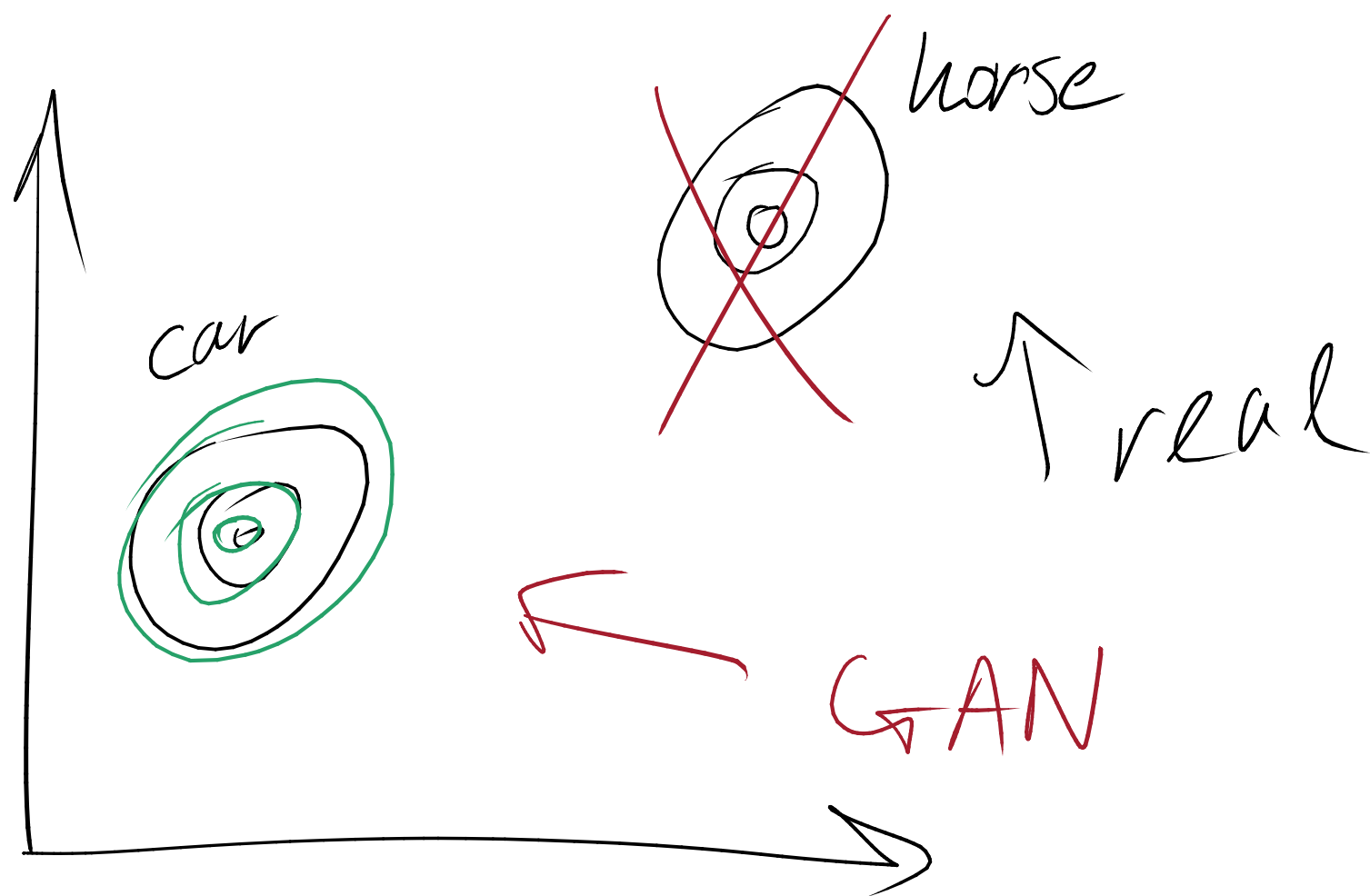


GAN:

mode collapse



VAE: Variational Auto Encoder

Auto encoders

"Manifold hypothesis"

data lives on a low-dimensional manifold.

Dimensionality reduction

$$x \in \mathbb{R}^D \rightarrow z \in \mathbb{R}^d \quad d < D \quad (d \ll D)$$

$$f(x) = z$$

* f - linear, $X \in \mathbb{R}^{N \times D}$

N - num of training samples

$$Z \in \mathbb{R}^{N \times d}$$

$$Z = X \Theta$$

$$\Theta \in \mathbb{R}^{D \times d}$$

Reconstruction

$$\hat{X} = Z \Phi$$

$$\Phi \in \mathbb{R}^{d \times D}$$

$$X \approx \hat{X} = X \Theta \Phi$$

$$MSE = \|X - \hat{X}\|_F^2 = \underbrace{\|X - X \Theta \Phi\|_F^2}_{\substack{\text{full} \\ \text{rank}}} \rightarrow \min_{\Theta, \Phi}$$

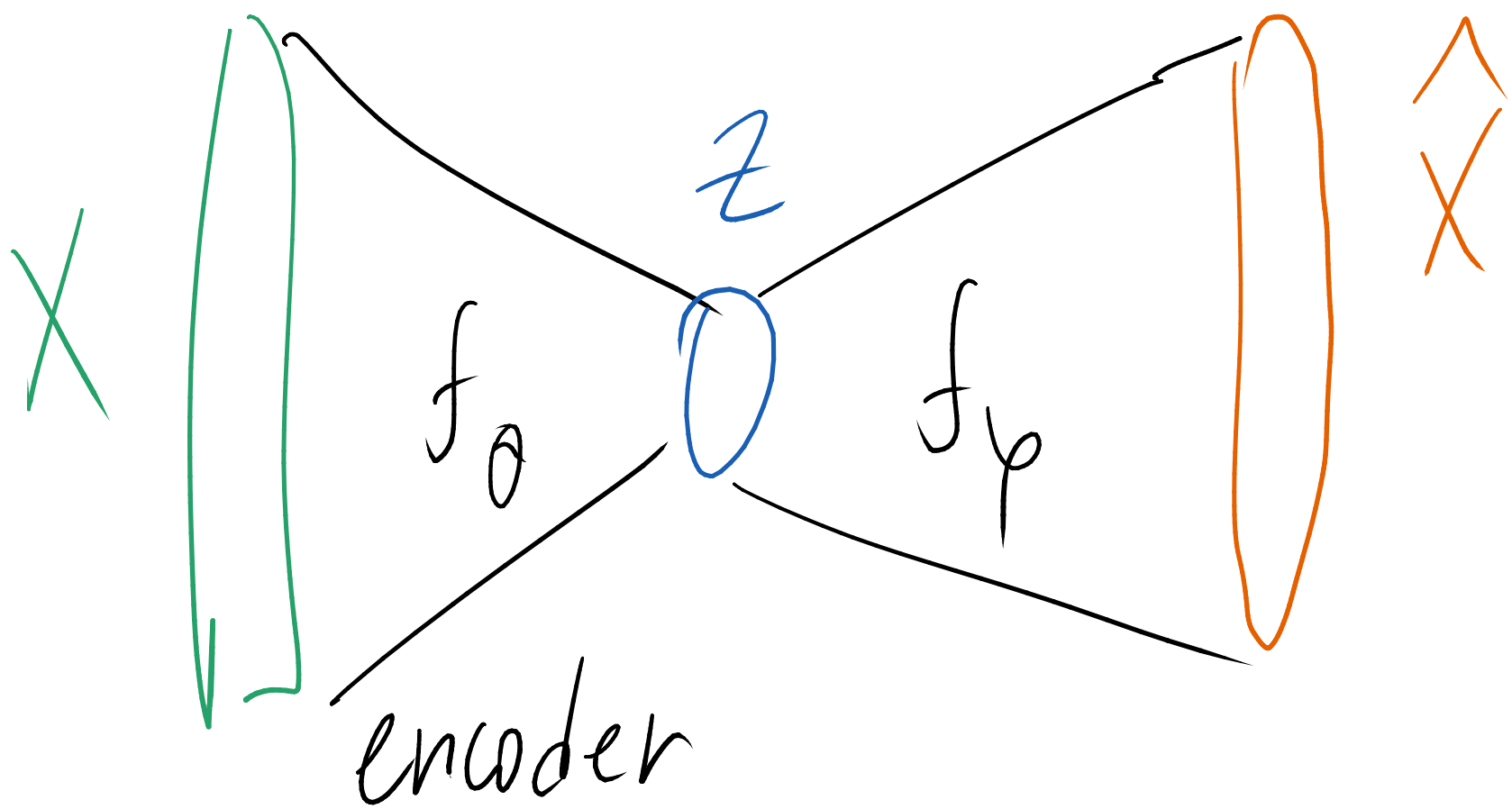
$$\|A\|_F = \sqrt{\sum_{ij} |A_{ij}|^2}$$

full
rank

$D \times d$ $d \times D$
rank d

— PCA (Principal Component Analysis)

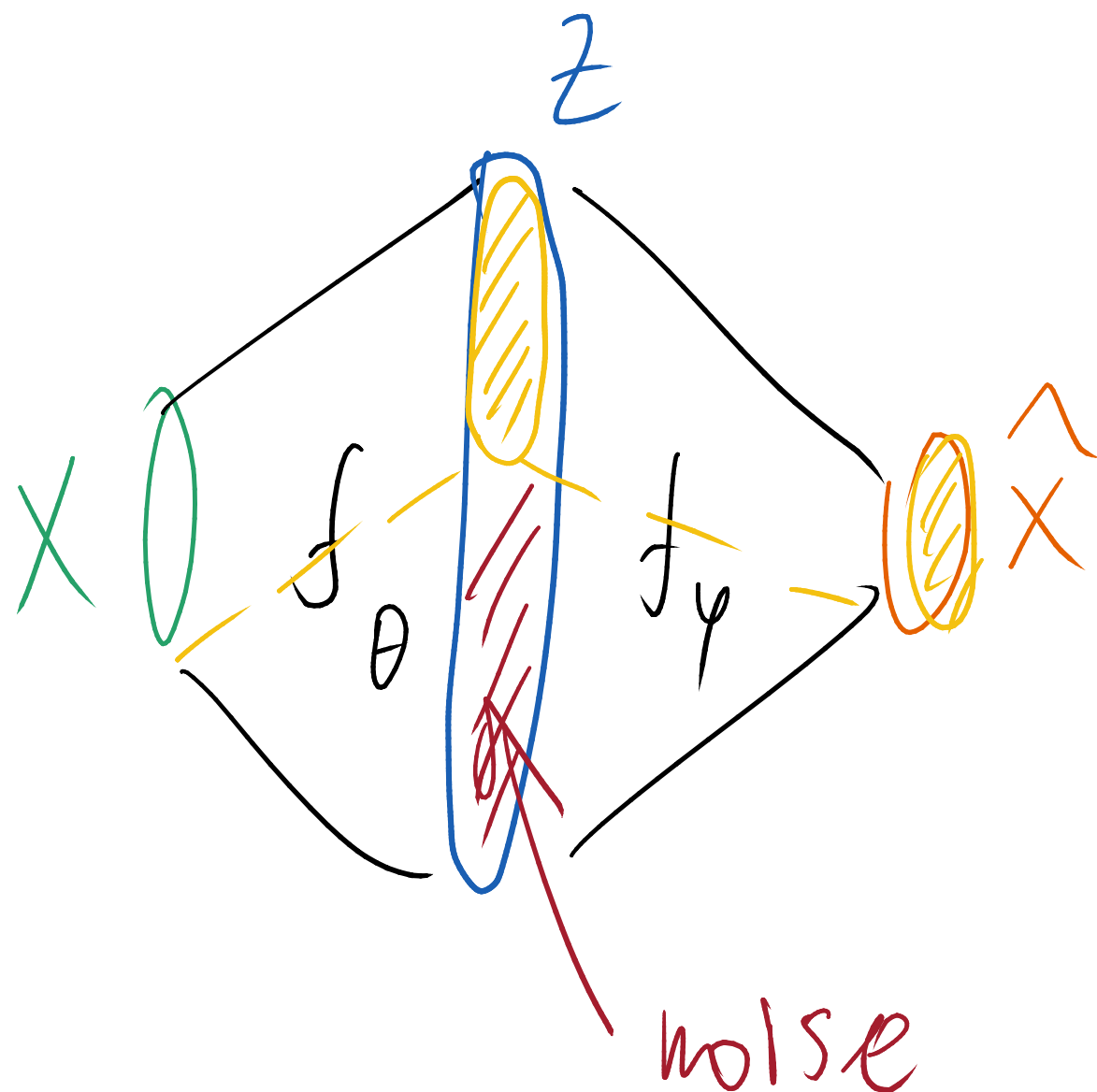
• f — non linear



p_{data}

$$\mathbb{E}_{x \sim p_{data}} \|x - \hat{x}\|_2^2 = \mathbb{E}_{x \sim p_{data}} \|x - f_{\phi}(f_{\theta}(x))\|_2^2$$

$\rightarrow \min$
 φ, θ
 $d > D$



Denoising Autoencoder:

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

p_{data}

$$X_{\text{noise}} = X + \epsilon \quad \epsilon\text{-noise}$$

$$\mathbb{E}_{X_{\text{noise}} \sim p_{\text{data}}} \|X_{\text{clean}} - f_{\phi}(f_{\theta}(X_{\text{noise}}))\|_2^2$$

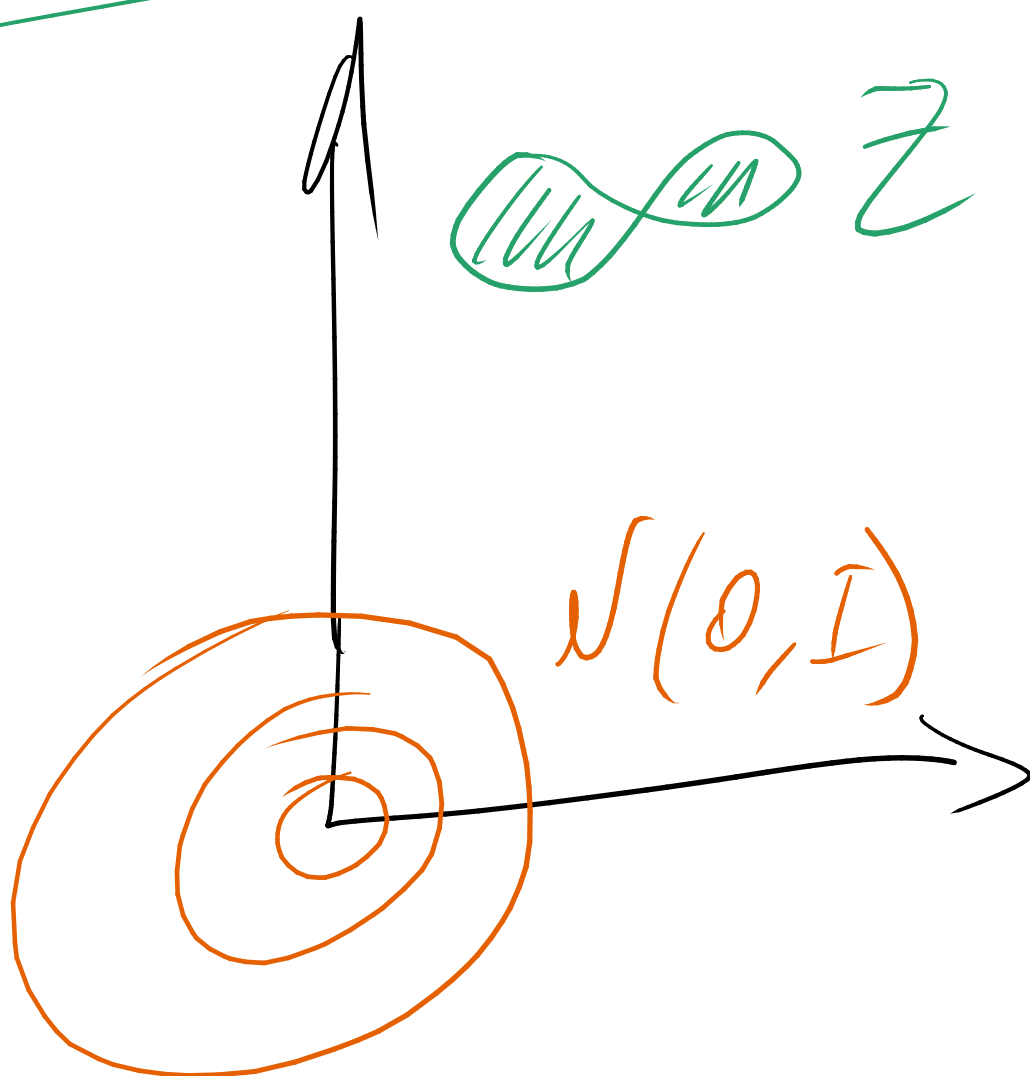
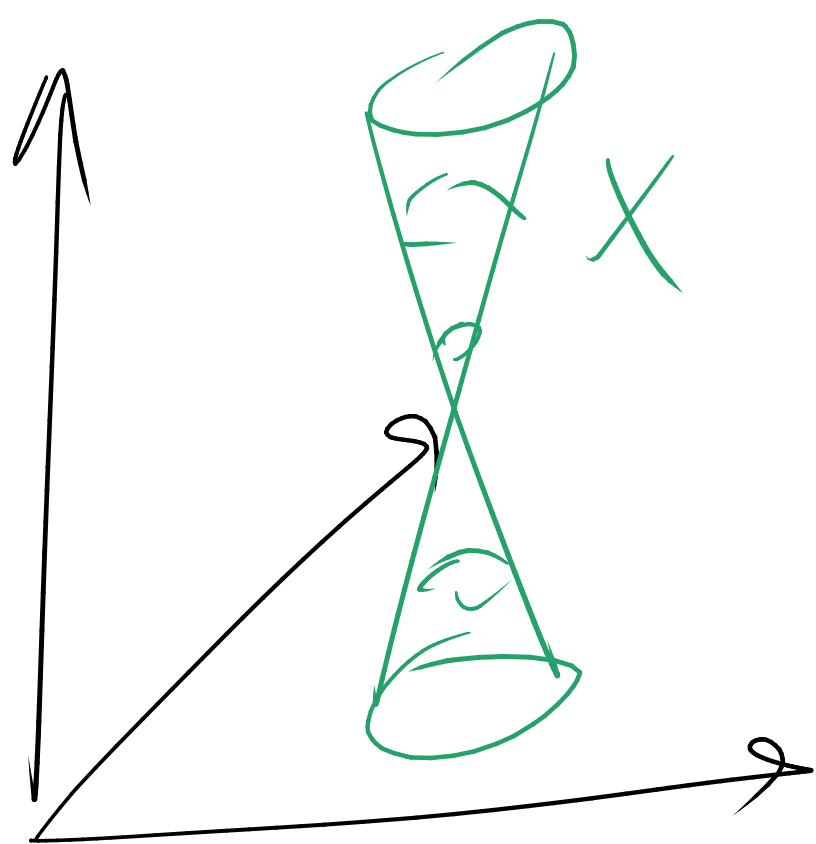
$\updownarrow$
 X_{clean}

$\rightarrow \min$
 φ, θ

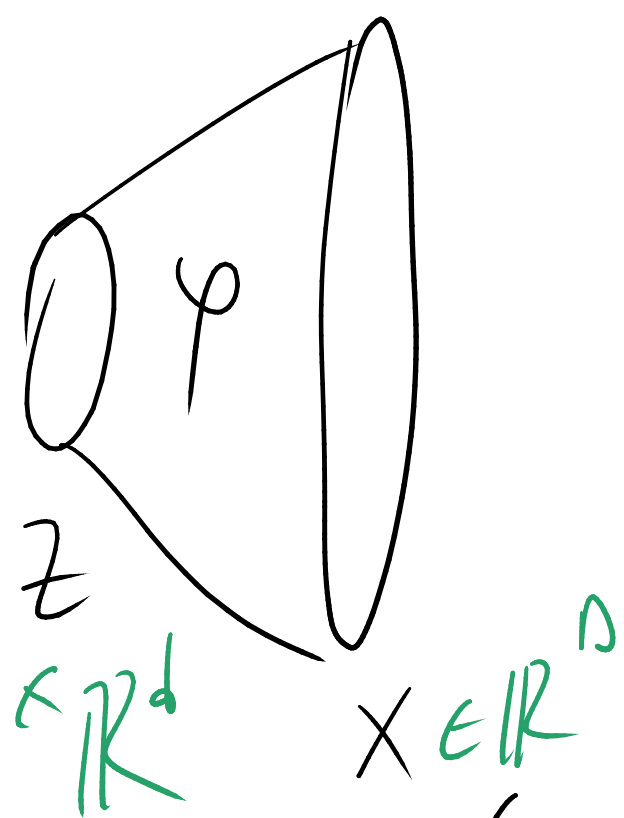
$$z \in \mathbb{R}^{128}$$

$$z \sim \mathcal{N}(0, I)$$

$$X_{\text{gen}} = f_{\phi}(z)$$



VAE



EM-algorithm

$$p(x, z | \psi) = p(x | z, \psi) p(z | \psi)$$

|| assume
 $p(z)$ prior

$$p(x | z, \psi) = \mathcal{N}(x | \mu_{\psi}(z), \Sigma_{\psi}(z))$$

$\in \mathbb{R}^D$ $\in \mathbb{R}^{D \times D}$

$$p(z) = \mathcal{N}(z | 0, I)$$

MLE: x - observed in pdata z - latent

$$\log p(x | \psi) = \log \int p(x, z | \psi) dz \rightarrow \max_{\psi}$$

Evidence Lower Bound (ELBO):

$$\begin{aligned} \log p(x | \psi) &= \log p(x | \psi) \cdot 1 = \log p(x | \psi) \cdot \int q(z) dz = \log p(x | \psi) \int q(z) dz \\ &= \log p(x | \psi) \int q(z) dz = \int q(z) \log p(x | \psi) dz = \int q(z) \log p(x, z | \psi) dz \\ &= \int q(z) \log p(x, z | \psi) dz = \int q(z) \log p(z | x, \psi) p(x | \psi) dz \end{aligned}$$

arbitrary distr. over z

independ. of z

$$= \int q(z) \log \frac{p(x, z | \varphi)}{p(z | x, \varphi)} dz =$$

$$= \int q(z) \log \frac{p(x, z | \varphi) q(z)}{p(z | x, \varphi) q(z)} dz =$$

$$= \int q(z) \log \frac{p(x, z | \varphi)}{q(z)} dz + \int q(z) \log \frac{q(z)}{p(z | x, \varphi)} dz =$$

$$= \underbrace{\mathcal{L}(q, \varphi, x)}_{\text{ELBO}} + \text{KL}(q(z) \parallel \underset{\substack{\uparrow \\ \text{posterior}}}{p(z | x, \varphi)})$$

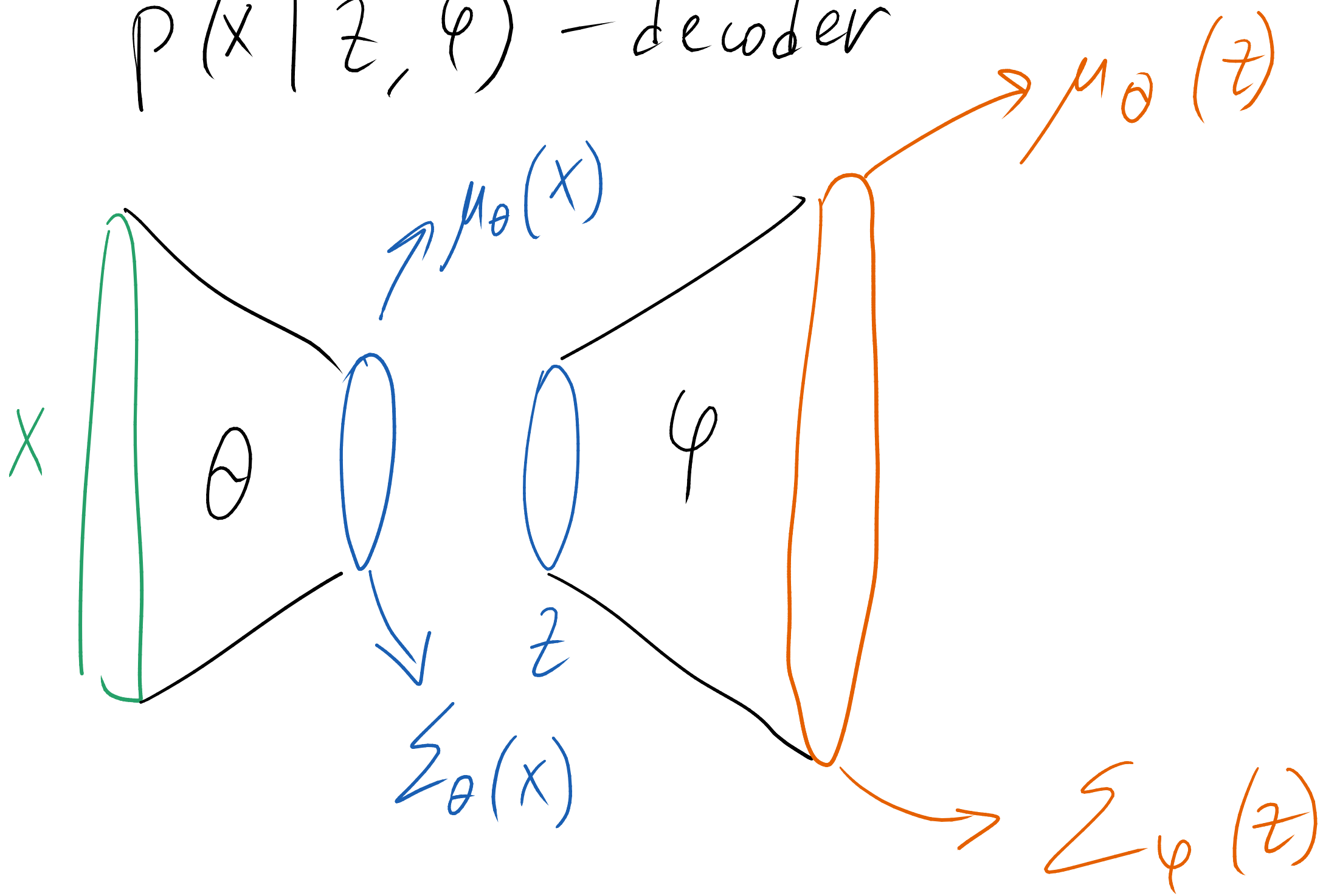
$$q = p(z | x, \varphi) - \text{posterior} \quad \text{KL} = 0$$

$$\mathcal{L}(q, \varphi, x) = \log p(x | \varphi)$$

$$\text{ELBO} \rightarrow \max \varphi$$

$$q(z) := q(z | x, \theta) \quad q\text{-encoder}$$

$p(x|z, \varphi)$ - decoder



$$q(z|x, \theta) = \mathcal{N}(z | \mu_{\theta}(x), \Sigma_{\theta}(x))$$

$$\mu_{\theta}(x) \in \mathbb{R}^d \quad \mu_{\varphi}(z) \in \mathbb{R}^D$$

$$\Sigma_{\theta}(x) = \text{diag}(\sigma_{\theta_1}^2(x), \dots, \sigma_{\theta_d}^2(x))$$

$$\Sigma_{\varphi}(z) = \text{diag}(\sigma_{\varphi_1}^2(z), \dots, \sigma_{\varphi_D}^2(z))$$

$$(\Sigma_{\varphi}(z) = I)$$

$$\mathcal{L}(\varphi, \theta, x) = \int q(z|x, \theta) \log \frac{p(x, z|\varphi)}{q(z|x, \theta)} dz \rightarrow \max_{\theta, \varphi}$$

$$\mathcal{L}(\varphi, \theta, x) = \int p(x, z | \varphi) = p(x | z, \varphi) p(z) dz =$$

$$\hat{x} = \mu_{\varphi}(z)$$

$$= \int q(z | x, \theta) \log \frac{p(x | z, \varphi) p(z)}{q(z | x, \theta)} dz =$$

$$= \int q(z | x, \theta) \log p(x | z, \varphi) dz +$$

$$+ \int q(z | x, \theta) \log \frac{p(z)}{q(z | x, \theta)} dz =$$

$$= \underbrace{\mathbb{E}_{q(z|x, \theta)} \log p(x | z, \varphi)}_{\text{MSE (reconstruction loss)}} - \underbrace{\text{KL}(q(z|x, \theta) \| p(z))}_{\text{VAE regularization}}$$

$$\nabla_{\theta} \mathbb{E}_{q(z|x, \theta)} \log p(x | z, \varphi) \approx \nabla_{\theta} \frac{1}{B} \sum_{i=1}^B \log p(x_i | z, \varphi)$$

$z \sim q(z | x_i, \theta)$

Reparameterization trick

$$z \sim \mathcal{N}(z | \mu, \Sigma^2) \quad \varepsilon \sim \mathcal{N}(\varepsilon | 0, I)$$

$$z = \mu + \Sigma \varepsilon$$

$$z \sim \mathcal{N}(z | \mu_\theta(x), \Sigma_\theta(x))$$

$$\varepsilon \sim \mathcal{N}(\varepsilon | 0, I) \quad z = \mu_\theta(x) + \Sigma_\theta(x)^{\frac{1}{2}} \cdot \varepsilon =$$

$$= \mu_\theta(x) + \sigma(x) \odot \varepsilon$$

$$\nabla_\theta \mathbb{E}_{q(z|x, \theta)} \log p(x|z, \varphi) =$$

$$= \nabla_\theta \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I)} \log p(x|z, \varphi) = z(\mu_\theta(x), \Sigma_\theta(x), \varepsilon)$$

$$\log p(x|z, \varphi) = \log \left[\frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \right]$$

$$x \in \mathbb{R}^D$$

$$z \in \mathbb{R}^D$$

$$\cdot \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right) =$$

$$= -\frac{D}{2} \log 2\pi - \frac{1}{2} \log |\Sigma| + (x-\mu)^T \Sigma^{-1} (x-\mu) =$$

const

$$= -\frac{1}{2} \sum_{i=1}^D \log \sigma_i^2 + \sum_{i=1}^D \frac{(x_i - \mu_i)^2}{\sigma_i^2} \propto \left\{ \Sigma_p(z) = I \right\}$$

$\propto \sum_{i=1}^D (x_i - \mu_i)^2$

$$q(z|x, \theta) = \mathcal{N}(z | \mu_\theta, \Sigma_\theta) \quad p(z) = \mathcal{N}(z | 0, I)$$

$$z_1 \sim \mathcal{N}(z_1 | \mu_1, \Sigma_1) \quad z_2 \sim \mathcal{N}(z_2 | \mu_2, \Sigma_2)$$

$$\text{KL}(z_1 \| z_2) = \frac{1}{2} \left[\text{tr}(\Sigma_2^{-1} \Sigma_1) - d + \right. \\ \left. + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) + \log \frac{|\Sigma_2|}{|\Sigma_1|} \right]$$

$$\text{KL}(q \| p) = \frac{1}{2} \left[\text{tr} \Sigma_\theta - d + \|\mu_\theta\|_2^2 - \log |\Sigma_\theta| \right] =$$

$$= \frac{1}{2} \sum_{i=1}^d (\sigma_i^2 + \mu_i^2 - 2 \log \sigma_i) - \frac{d}{2}$$

$$\sigma_i^2 = \exp(\log \sigma_i^2) \quad \log \sigma^2 - \text{output of head}$$

VAE GAN — all the downsides